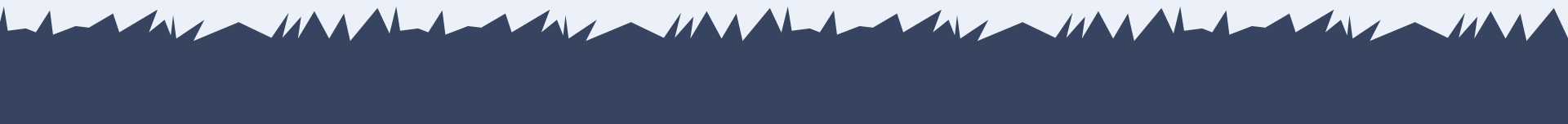
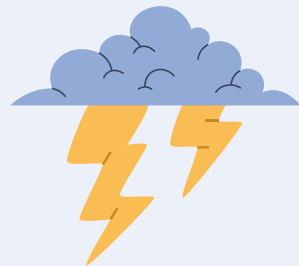


Rainfall by the Numbers

By: Mia Parker, Emily Strickland, and Aurora Hayden



Our Data



Description

We used a weather dataset from Kaggle containing daily observations across Australia. It includes temperature, rainfall, wind, pressure, and humidity data used to analyze weather patterns.

Variables

Target: Rainfall (mm)

Predictors: Humidity9am, RainToday, WindDir9am, MaxTemp, Location

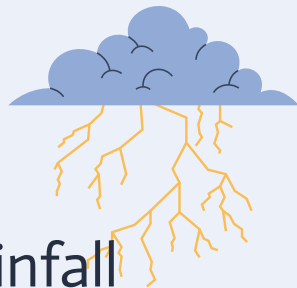
Data Type: Numeric and Categorical

Timeframe

From November, 2007 to June, 2017.



Introduction/First Thoughts



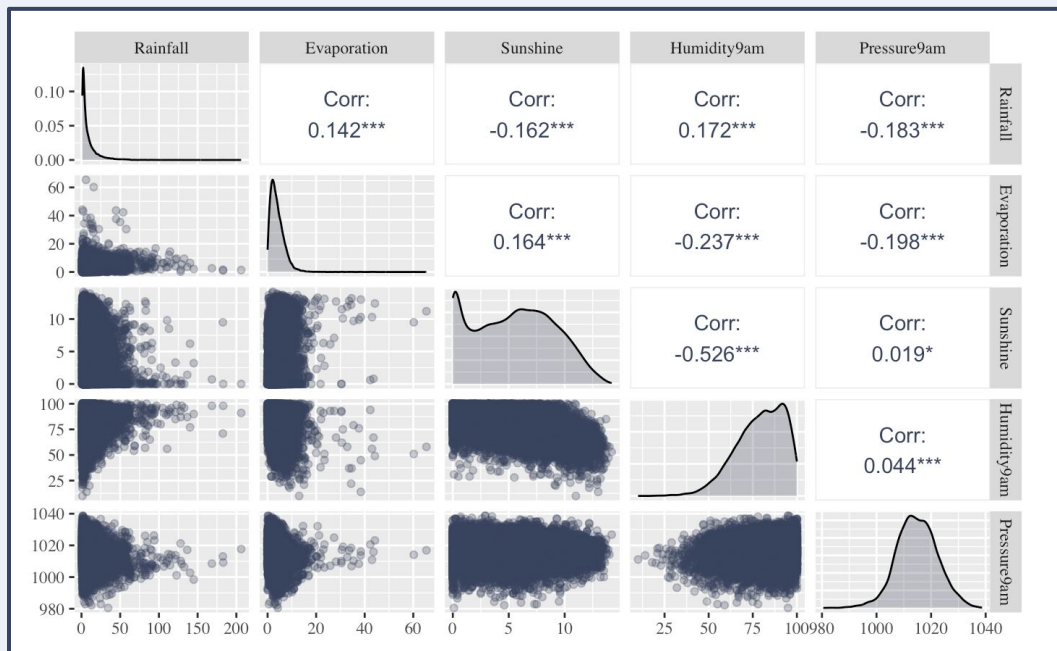
- We were curious about how well we could predict rainfall based on morning weather conditions.
- The dataset gave us a lot of variables to explore and compare across models.
- We decided to try different regression models to see which one performed best.





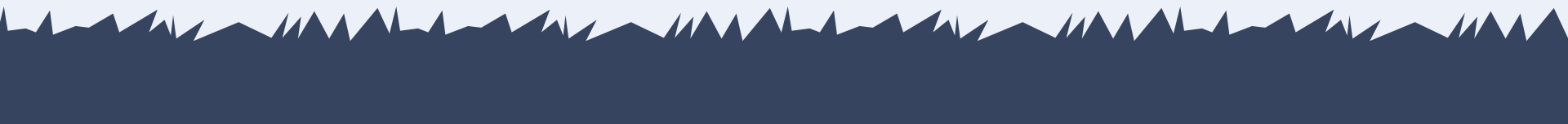
Exploratory Data Analysis

- GGpairs of different variables within the data
- Quick method to see relationships in the data
- Can see that Pressure 9am and Humidity 9am have the highest correlation with Rainfall



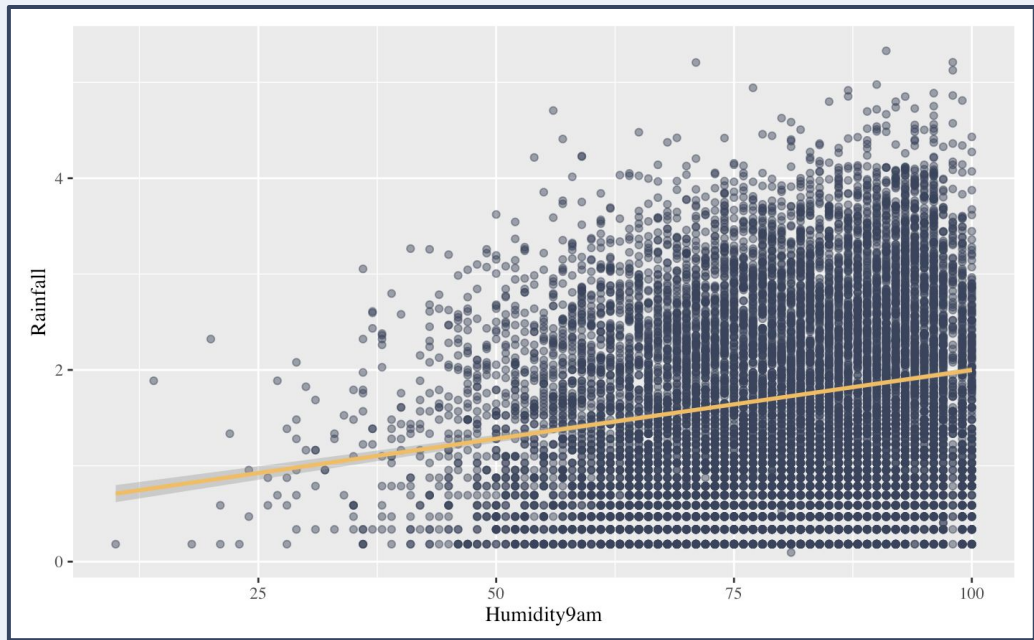
Question 1:

**Can we predict the
amount of rainfall using
our data?**





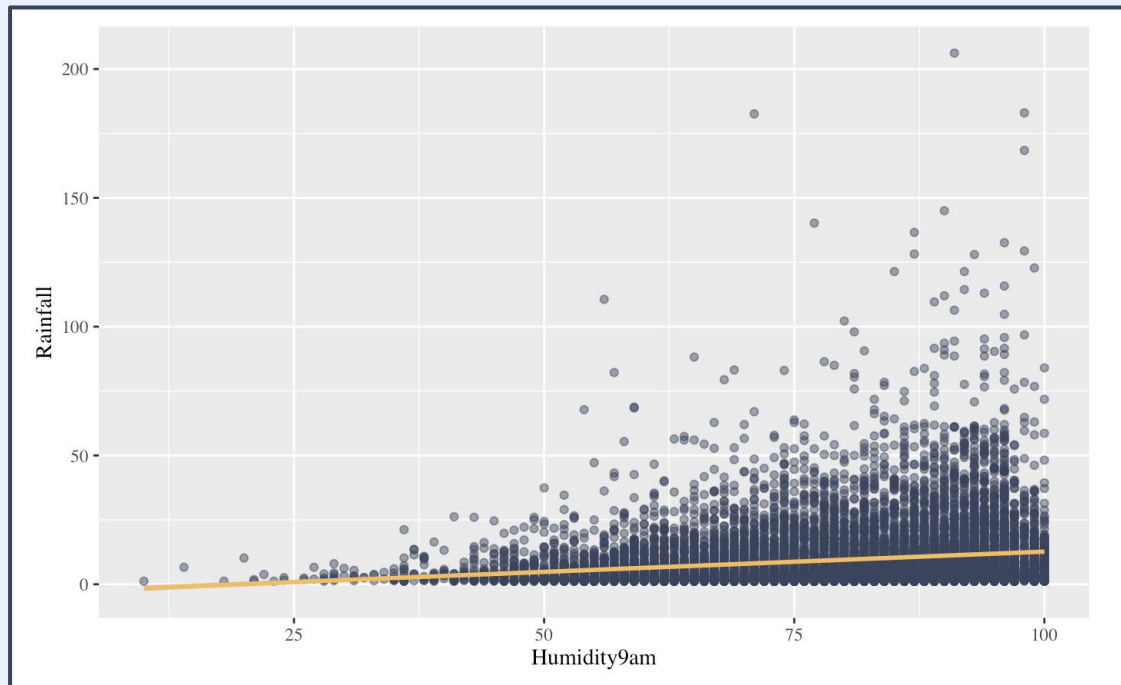
Linear Regression with Log



Observations:

- We used simple linear regression to model rainfall based on Humidity at 9 AM.
- The line of best fit shows a positive relationship: as humidity increases, so does rainfall.
- While there is a clear trend, the spread of points shows that humidity alone is not enough for strong predictions.

Linear Regression without Log()



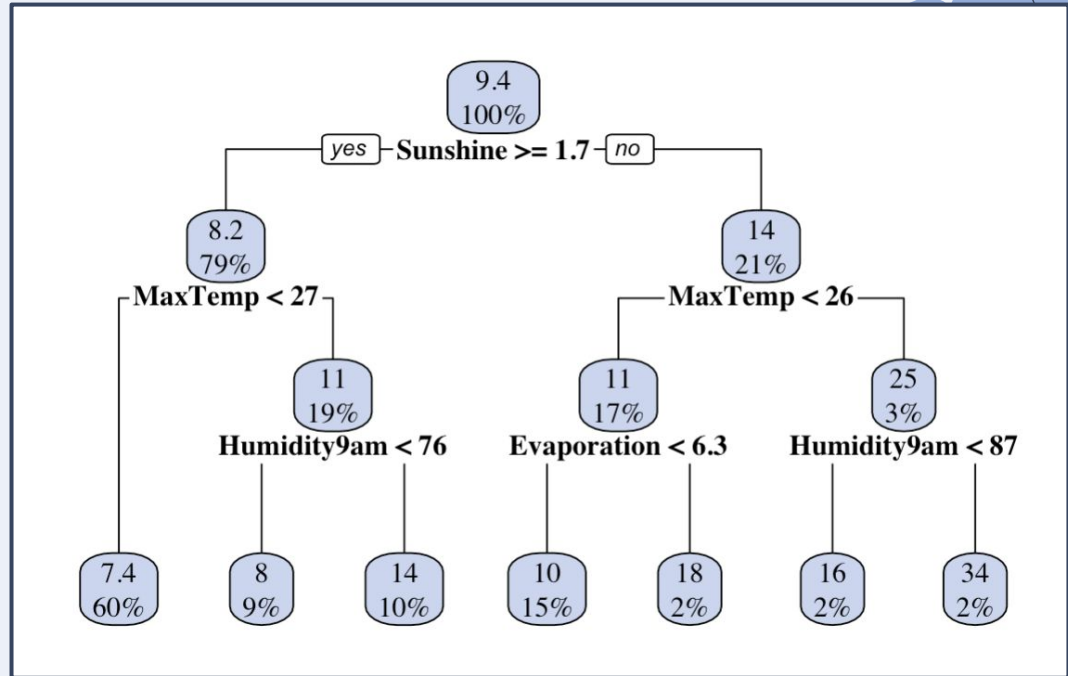
Observations:

- Data is dense and grouped around the x axis
- Our line of best fit is strongly affected by these low values
- Higher rainfall days have little effect on our model



Tree for Rainfall

- Decision Tree that estimates amount of rainfall using other variables
- Higher amounts of sunshine leads to less rain
- Higher humidity leads to more rain



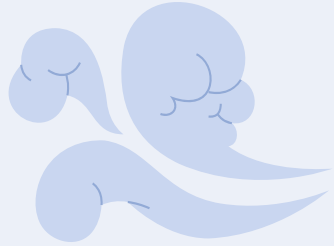
RMSE Comparison



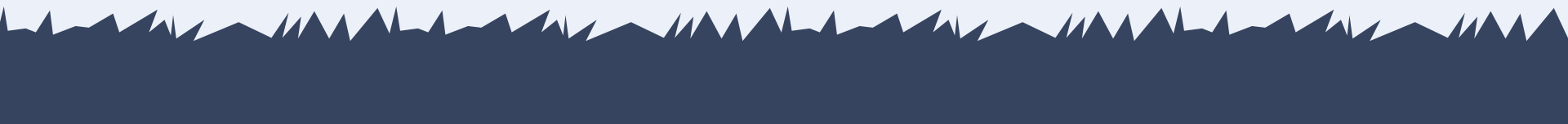
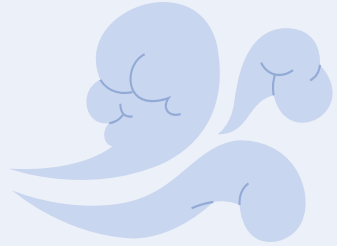
Type	RMSE
Linear Regression Log	5.40
Linear Regression	6.45
Regression Tree	6.65

Observations:

- RMSE stands for Root Mean Squared Error - a measure of how far off our predictions are from actual values.
 - A lower RMSE means better predictive accuracy.
 - The Log of Linear Regression had the lowest RMSE, meaning it performed slightly better than the other models.
- 

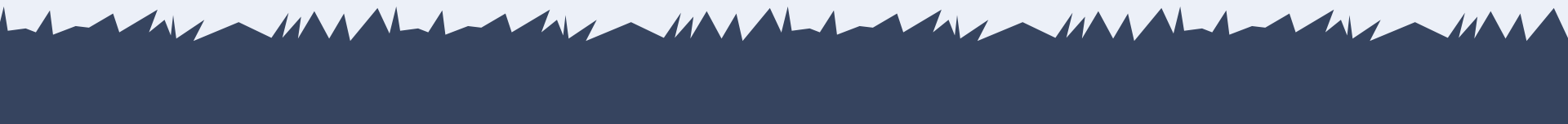


Pivoting Focus

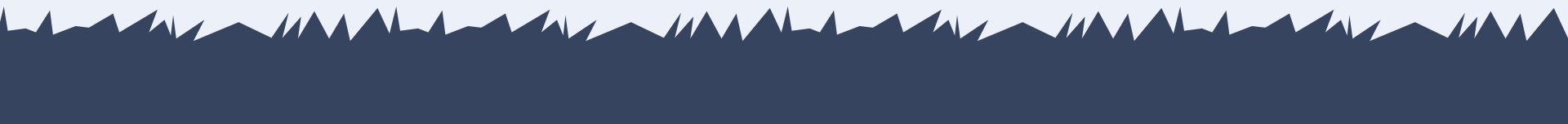
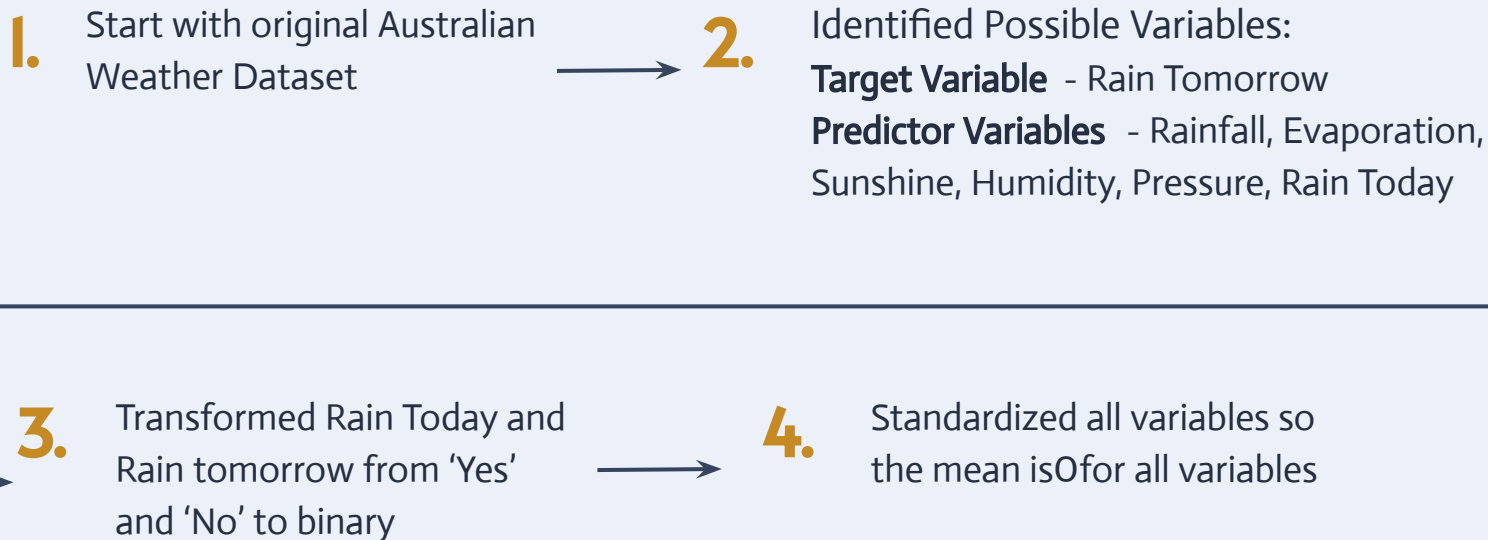


Question 2:

**Can we predict whether
it will rain tomorrow
using our data?**



Variables and Process





Predicting Next Day Rain

What did we make?

- Called a KNN Algorithm
- Stratifies the data using the variables we gave it
- Correct predicted values are displayed on the diagonal

Goal: to accurately predict the if it will or will not rain tomorrow

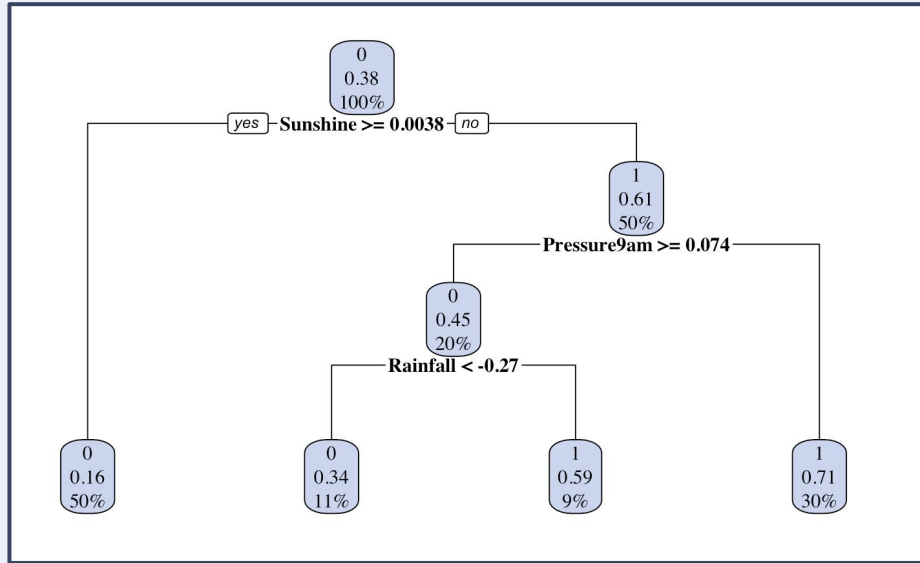
Observations

- Our model predicts that it won't rain more accurately than it will rain
- We have an accuracy rate of **0.7603** which means it is correct 76% of the time

		Actual	
		Predicted 0	Predicted 1
Actual	0	4978	1152
	1	1022	2577



Tree from Model



What did we make?

- A classification tree
- Shows how our data is sectioned by the variables given

Keep in Mind: Our variables are standardized around 0 so displayed numbers are not actual values

Observations

- Given 6 variables, the model chose to use these three
- Days with more sunshine are likely to not rain tomorrow
- Higher amounts of rainfall today correlate with rain tomorrow



Conclusion and Analysis

Question 1:

- Log of linear regression captures data better than normal linear regression
- The decision tree has a worst accuracy
- Low temp and high humidity lead to the highest amount of rain

Question 2:

- The best predictors for if it will rain tomorrow are sunshine, pressure and rainfall
- We can do this with an accuracy of 76%
- We can identify more sunshine and less rain correlate with no rain tomorrow





Thank you for listening!

Any Questions?

